

Euclid's Elements — Book I

Proposition 28

Formal Reconstruction — Technical Documentation

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Further angle criteria for parallel lines

1 Introduction

Proposition I.28 gives two sufficient angle conditions for parallelism. A transversal meets two straight lines. Parallelism follows either from an equality of corresponding angles or from the condition that the two interior angles on the same side are together equal to two right angles.

The central structural fact is that neither branch introduces a new parallelism mechanism. Both normalize their angle data to the alternate interior angle hypothesis of Proposition I.27:

$$\text{angle data} \longrightarrow \angle AGH \cong \angle GHD \longrightarrow \text{I.27} \longrightarrow AB \parallel CD.$$

This direction is neutral. The current production Lean file requires only incidence and congruence, not `HilbertEuclideanPlane`. The logical reversal occurs in Proposition I.29.

2 Classical Configuration

Let the straight line EF meet AB at G and CD at H . The common geometric configuration is

$$A - G - B, \quad C - H - D, \quad E - G - H - F.$$

The two clauses of the proposition differ only in the angle condition placed on this same configuration.

2.1 Corresponding-angle criterion

Assume

$$\angle EGB \cong \angle GHD.$$

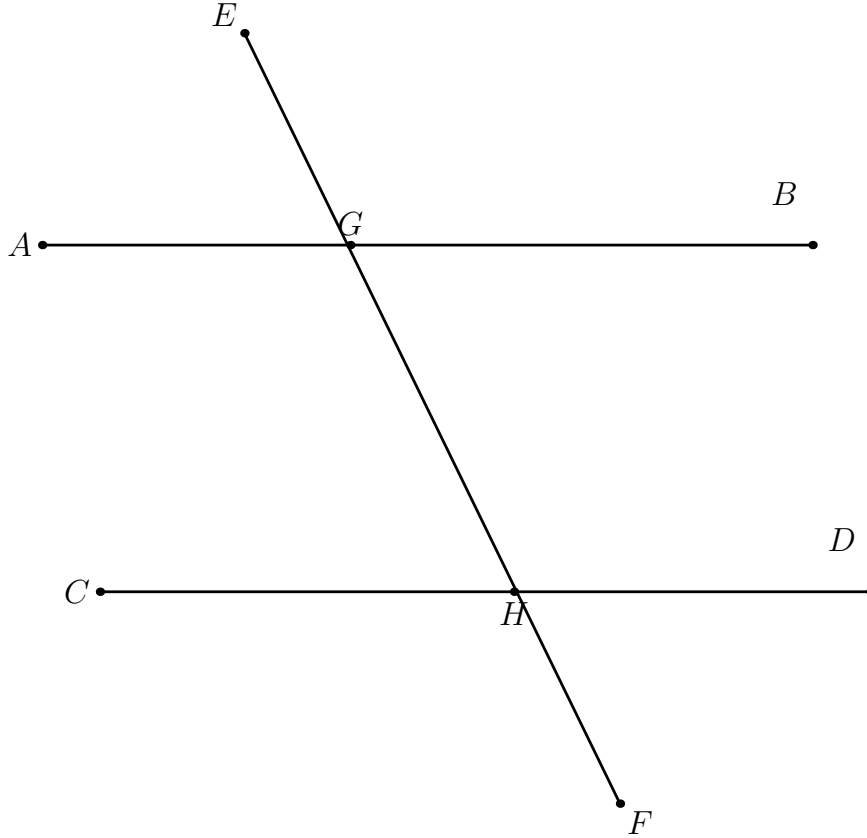


Figure 1: The common transversal configuration for Proposition I.28. The selected arms through A and D lie on opposite sides of the transversal in the explicit formal statement. No angle equality and no final parallelism conclusion is encoded in the drawing; the two clauses of I.28 are different hypotheses on this same geometric state.

Since $A - G - B$ and $E - G - H$, the angles $\angle EGB$ and $\angle AGH$ are vertical. Proposition I.15 gives

$$\angle AGH \cong \angle EGB.$$

By transitivity,

$$\angle AGH \cong \angle GHD,$$

and Proposition I.27 yields

$$AB \parallel CD.$$

Thus the classical dependency chain of the first clause is

$$\text{corresponding angles} \longrightarrow \text{I.15} \longrightarrow \text{alternate angles} \longrightarrow \text{I.27}.$$

2.2 Same-side interior-angle criterion

Assume instead that

$$\angle BGH \quad \text{and} \quad \angle GHD$$

are together equal to two right angles. Because $A - G - B$, the angles $\angle AGH$ and $\angle BGH$ are supplementary. Euclid uses Proposition I.13 and Common Notion 3 to cancel the common angle and obtain

$$\angle AGH \cong \angle GHD.$$

Again I.27 gives $AB \parallel CD$. The classical dependency chain is therefore

$$\text{I.13} + \text{C.N.3} \longrightarrow \text{alternate angles} \longrightarrow \text{I.27}.$$

3 Current Lean Architecture

The production module imports only `Proposition27` and exposes two public theorem entrypoints:

```
euclid_proposition_28_corresponding
euclid_proposition_28_same_side
```

There is no proposition-local helper theorem and no geometric case split. The two theorem entrypoints formalize the two alternative clauses of the classical proposition over the same transversal geometry.

Both statements use the explicit line `trans` through G, H and include the orientation condition

```
HilbertOppositeSide Geo A D trans
```

needed by the explicit transversal form of Proposition I.27.

3.1 Corresponding-angle branch

The first theorem receives

$$A - G - B, \quad C - H - D, \quad E - G - H, \quad \angle EGB \cong \angle GHD.$$

The current proof first reverses the transversal betweenness to obtain $H - G - E$. It then invokes

```
VerticalAngles
```

to prove a congruence equivalent to

$$\angle AGH \cong \angle EGB.$$

After endpoint reversal on the second angle and transitivity with the hypothesis, the exact alternate-angle input is obtained:

$$\angle AGH \cong \angle GHD.$$

The branch ends by calling

```
euclid_proposition_27_transversal
```

with the same intersection points G, H , the same transversal, and the same opposite-side orientation.

Schematically,

$$\begin{aligned} & \angle EGB \cong \angle GHD \\ & \downarrow \text{VerticalAngles} \\ & \angle AGH \cong \angle EGB \\ & \downarrow \text{transitivity} \\ & \angle AGH \cong \angle GHD \\ & \downarrow \text{euclid_proposition_27_transversal} \\ & AB \parallel CD. \end{aligned}$$

3.2 Same-side branch

The second theorem does not introduce numerical angle measure or an angle addition operation. Its hypothesis is the synthetic predicate

`HilbertAnglesEqualTwoRightAnglesWithSupplement`
`Geo G H D H G B A`

which supplies directly

$$B - G - A \quad \text{and} \quad \angle GHD \cong \angle HGA.$$

Betweenness symmetry recovers $A - G - B$. Symmetry and endpoint reversal of angle congruence then normalize the second component to

$$\angle AGH \cong \angle GHD.$$

The proof again terminates in `euclid_proposition_27_transversal`.

Hence the actual formal branch is

$$\begin{array}{c} \text{HilbertAnglesEqualTwoRightAnglesWithSupplement} \\ \downarrow \\ B - G - A, \quad \angle GHD \cong \angle HGA \\ \downarrow \text{ symmetry/reversal } \\ \angle AGH \cong \angle GHD \\ \downarrow \text{ euclid_proposition_27_transversal } \\ AB \parallel CD. \end{array}$$

The classical use of I.13 and Common Notion 3 is therefore represented in Lean by the shape of the synthetic supplementary-angle hypothesis rather than replayed as angle arithmetic inside this theorem.

4 Downstream Geometry in Proposition I.27

Both I.28 branches delegate their final geometric work to the explicit transversal theorem of Proposition I.27. That theorem chooses an interior point between its two transversal intersections, obtains two same-ray relations, normalizes the alternate angles, applies the neutral line-level alternate-angle criterion, and transports the resulting local parallelism back to the original carriers.

After substituting the I.28 intersection points, this hidden point lies between G and H . It is not constructed by either I.28 theorem itself. Consequently it belongs diagrammatically to the audit of I.27 and is not repeated as a new I.28 figure.

This distinction is useful: the direct architecture of I.28 is angle normalization followed by a call to I.27; the deeper geometry of the parallelism endpoint is a dependency, not a new state introduced by I.28.

5 Neutral Geometry

Both theorem statements require

[HilbertIncidence Geo]
[HilbertCongruence Geo]

and neither requires `HilbertEuclideanPlane`. Thus I.28 stays on the neutral side of the boundary:

angle condition $\longrightarrow$ parallel lines.

I.29 reverses this implication and is where the Euclidean parallel axiom enters the Book I sequence.

6 Diagrammatic Audit

The preserved Book1R chapter contained three separate floats:

1. the common transversal configuration;
2. the same configuration with arcs marking the corresponding-angle branch;
3. the same configuration with arcs marking the same-side branch.

Only the first records a geometric state. Figures 28.2 and 28.3 changed only which angles of the already drawn configuration were being discussed. They introduced no point, extension, intersection, betweenness relation, side relation, auxiliary line, or geometric case.

The audited chapter therefore keeps one figure only. The two angle criteria remain fully documented in the prose and proof DAGs, but they do not receive separate diagrams. This is an intentional application of the principle that a figure should record geometry rather than tactic state or a change of logical interpretation.

The formal orientation condition A, D on opposite sides of the transversal is stated in the caption and in the Lean architecture section. It does not require a second picture because it describes the same visible configuration.

There is also no geometric case split in the production file. The two clauses are two public theorem entrypts with different hypotheses, not two positional cases discovered inside one proof.

7 Conclusion

Proposition I.28 is a pair of neutral normalization theorems over one common transversal geometry. The corresponding-angle clause uses vertical angles; the same-side clause receives the supplementary relation directly in synthetic form. Both produce the same alternate-angle congruence and then delegate parallelism to Proposition I.27.

Diagrammatically, one common figure is therefore sufficient. The reduction from three historical floats to one audited figure removes duplicate pictures without suppressing any genuine geometric state.